

Visualisation and sea level rise

2016 AAPT Workshop June 13-14-15, 2016
"Sink or Swim: Planning for Climate Change"

Eric Rapaport

School of Planning



**DALHOUSIE
UNIVERSITY**



Visualisation

- **Landscape visualisations**

depict real places seen by people in the area, and help the viewer understand subject matter by visually illustrating significant characteristics and possible future conditions

Why use landscape visualization

- When **people see images of their community** integrated with scientific data drawn from maps, graphs, or charts, the information suddenly becomes real: the data's meaning becomes clear in the context of actual places that people recognize.

Sheppard, Stephen. "Local Climate Change Visioning: A New Process for Community Planning and Outreach using Visualization Tools." *Plan Canada*, 48:1 (Spring 2008): 36-40.

Engaging the Public

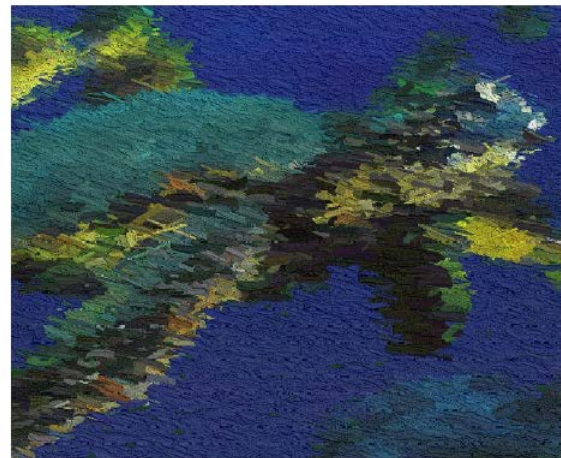
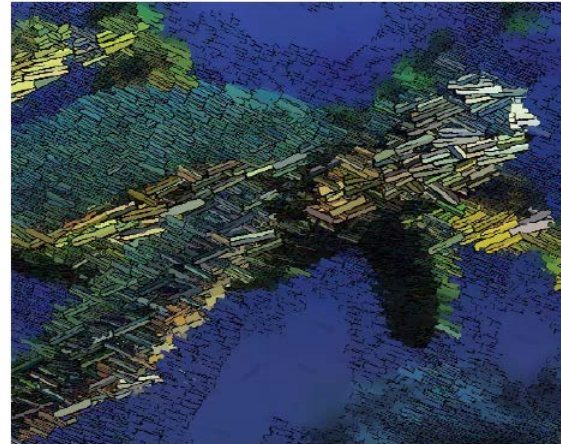
- Goals of Visualization
 - Foster exchange on information, 1 way and 2 way
 - Education
 - Affect Policy
 - Better decision making

Criteria for good landscape visualisation

- **Comprehension**
- **Representative**
- **Accuracy**
- **Credibility**
- **Defensible**
- **Engaging**
- **Accessible**

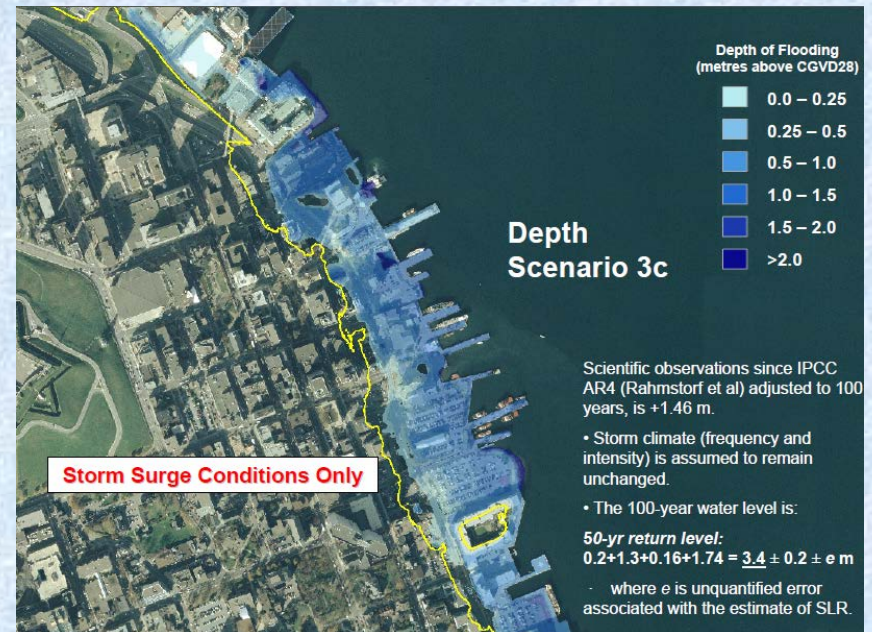
Comprehension

- Understandability of the visual material
- Local known context



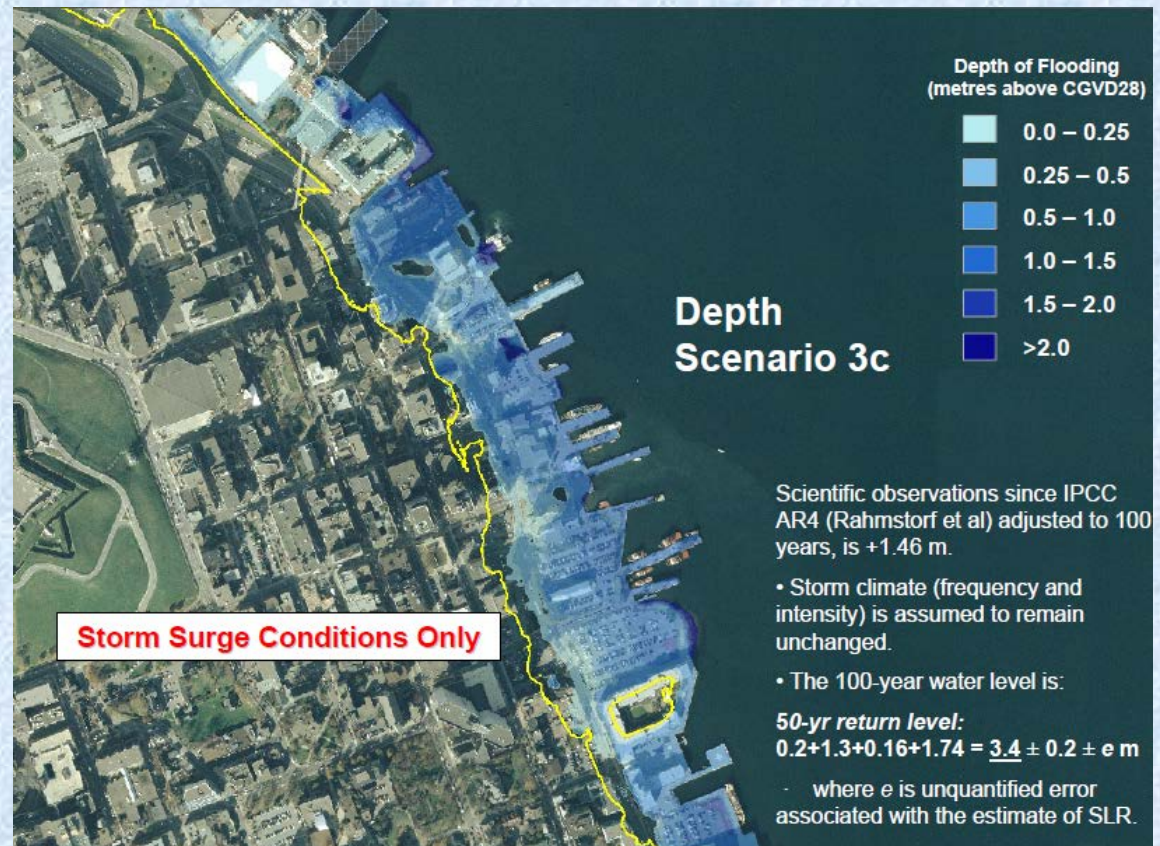
Representation

- Visualisations are representative if they contain:
 - Clear and distinctive views of the area
 - Views under different conditions
 - Images with sufficient scope or scale to understand the anticipated effect

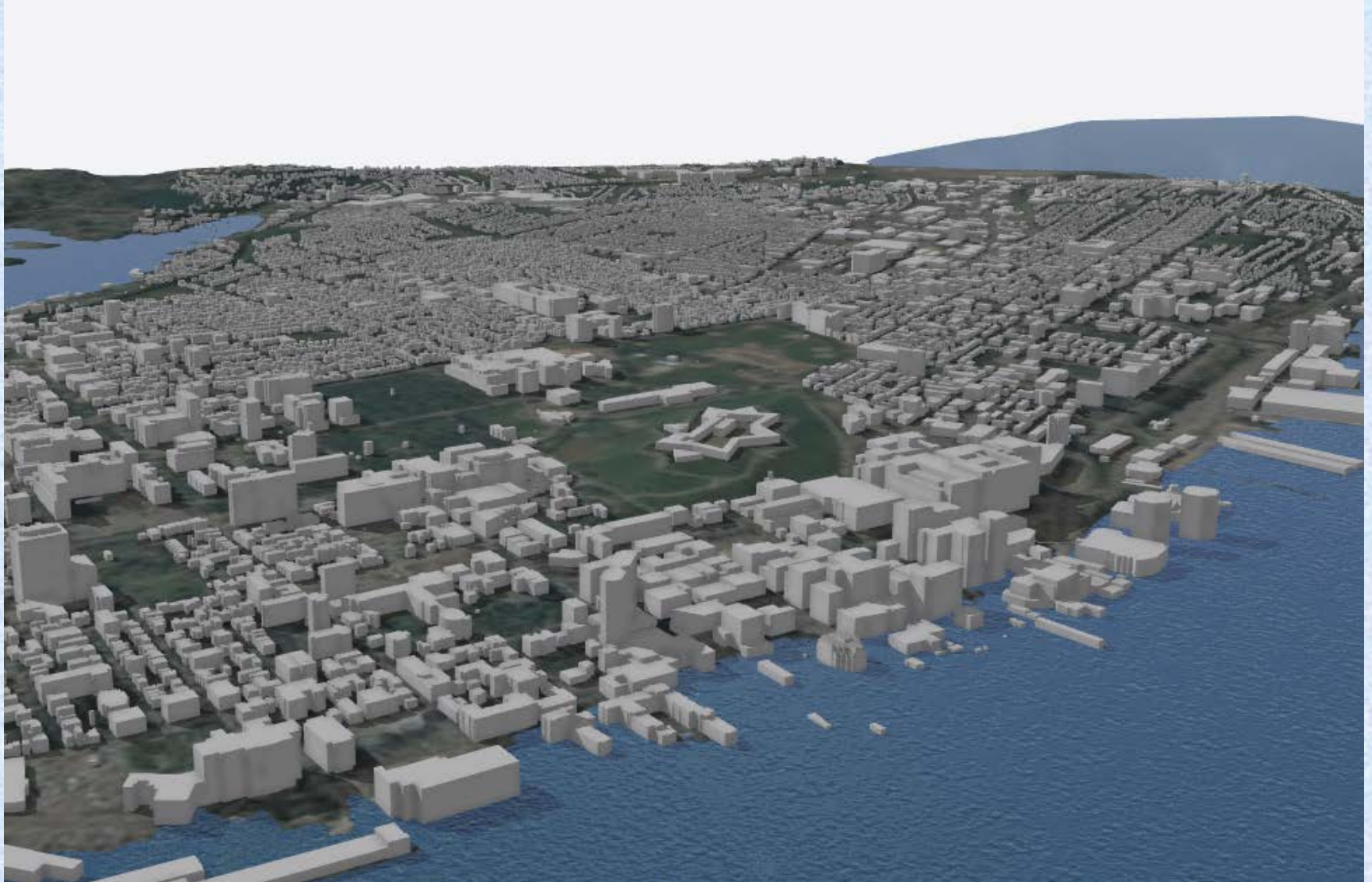


Accuracy

- Is there a correct representation of the current conditions that future conditions can be considered plausible?
 - Are these buildings underwater?



Accuracy



Credibility

- Trust and unbiased information

Scenario

Sea level rise (Scenario 2100)	0.57 m Regionalize IPCC 4th AIF1 *
Subsidence (fixed rate)	0.85 m Regionalize for the IPCC 5th RCP8.5 ++
Ocean Tides (fixed rate)	1.3 m alternative sea level rise prediction **
Storm returns (fixed rate)	• Why did the worse case scenario change from 0.57 to 0.85? • Why not used the <u>fixed rate</u> of relative SLR at the Halifax = 0.32 cm?

*Forbes,D. et at. 2009. Halifax harbour extreme water levels in the context of climate change: scenarios for a 100-year planning horizon. Geological Survey of Canada

++Daigle. R. 2014. Sea-Level Rise and Coastal Flooding Estimates for Chignecto Isthmus and Halifax Harbour Based on IPCC 5th Assessment Report. Nova Scotia Department of Environment

** Rahmstorf, S. 2007. A semi-empirical approach to projecting future sea-level rise. *Science*, 315, 368- 370.

Defensible

- The production method of the visualization needs to be clear and accessible information

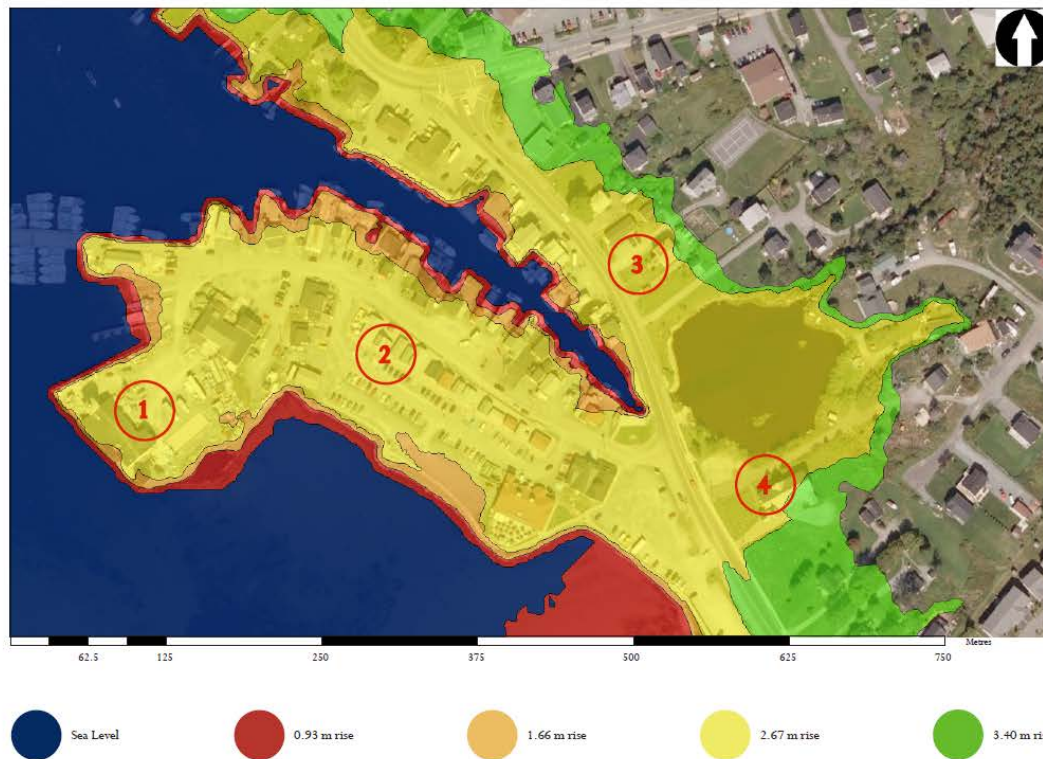


Photo by Florian Goetz Visualizing Future Extreme Water Levels in Halifax's Northwest Arm in 2100. Dalhousie University, School of Planning, PLAN 4500

Engaging

- Measure the ability of the visualisation material to maintain audience attention.

Hazard Risk Colors: Map and Photo Simulations

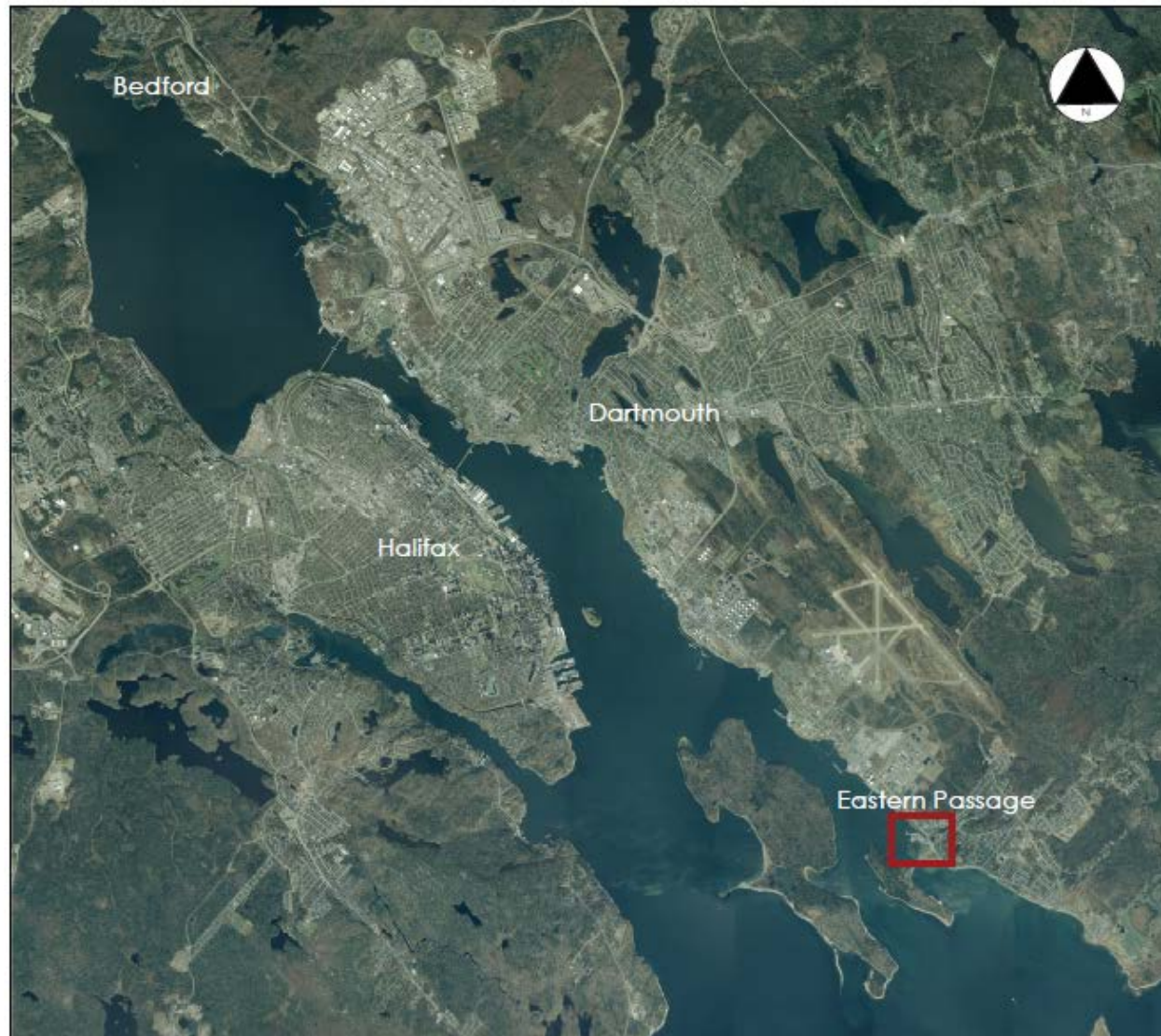


Accessible

- This refers to the physical and logistic ability of the audience to access the material.



Sea Level Rise and Communication



Source: GeoNova, 2003

1:80,000

Figure 1: Study Area in Relation to Halifax and Dartmouth

How to represent sea level rise and flooding scenarios

Maps



Renderings



Photo Simulations



Sea level rise and worst case flooding scenarios

Scenario 1: 0.73 metres

Scenario 2: 1.66 metres

Scenario 3: 2.67 metres

Scenario 4: 3.40 metres

Scenario 5: 4.4 metres

Forbes, D. et al. 2009. Halifax harbour extreme water levels in the context of climate change: scenarios for a 100-year planning horizon. Geological Survey of Canada

<https://www.halifax.ca/regionalplanning/documents/OF6346final.pdf>

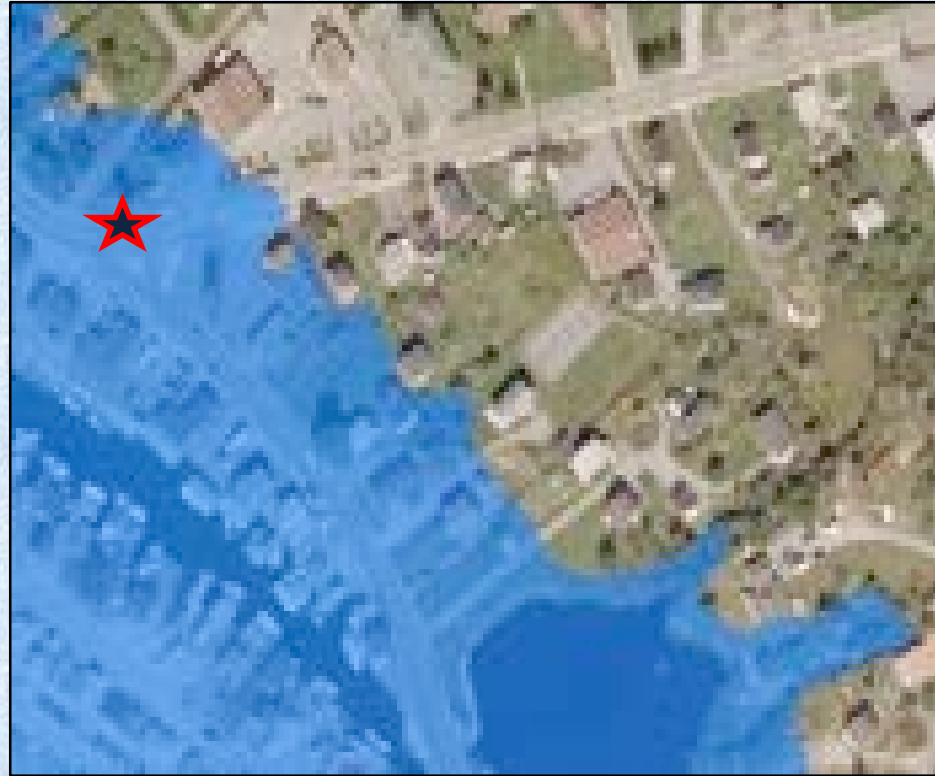
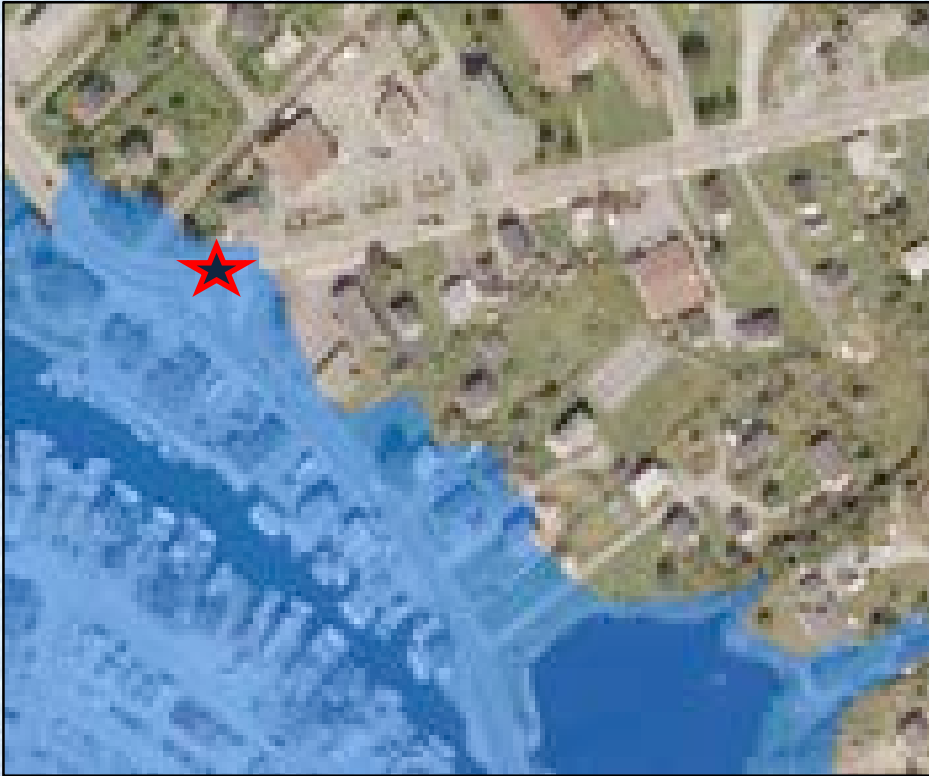
How to represent sea level rise and flooding scenarios

- Maps
- Rendering
- Photo simulations
- Computers and dynamic models
- Tactile models

Maps

Passive Observation – 1D

Realism and wide scope



Rendering

Passive Observation -2D

Abstract and limited scope

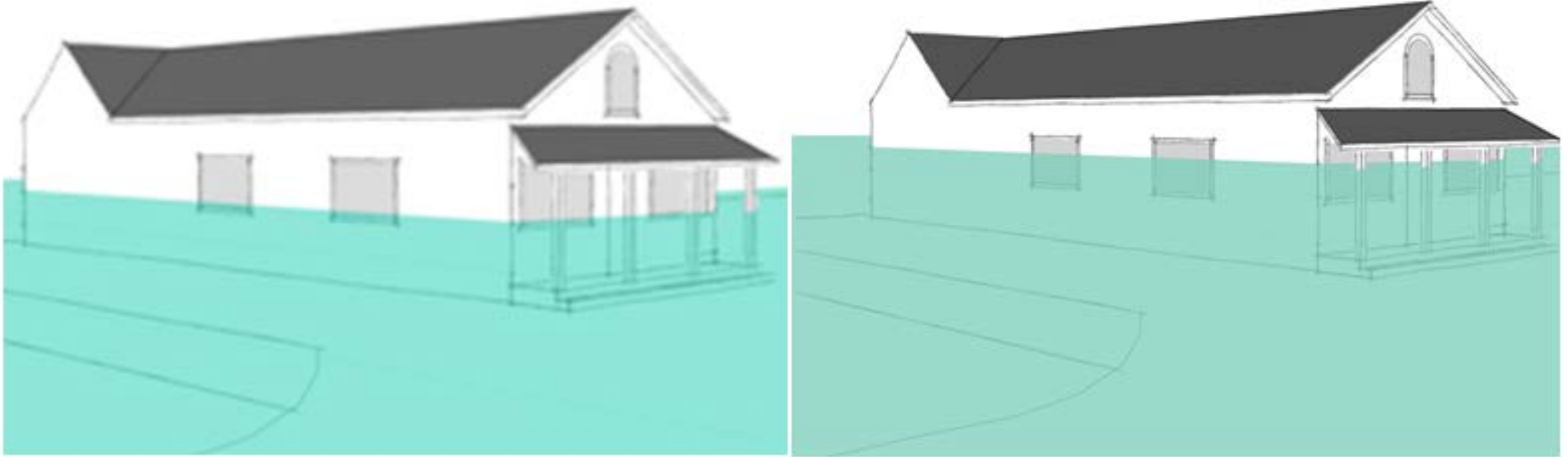


Photo Simulation

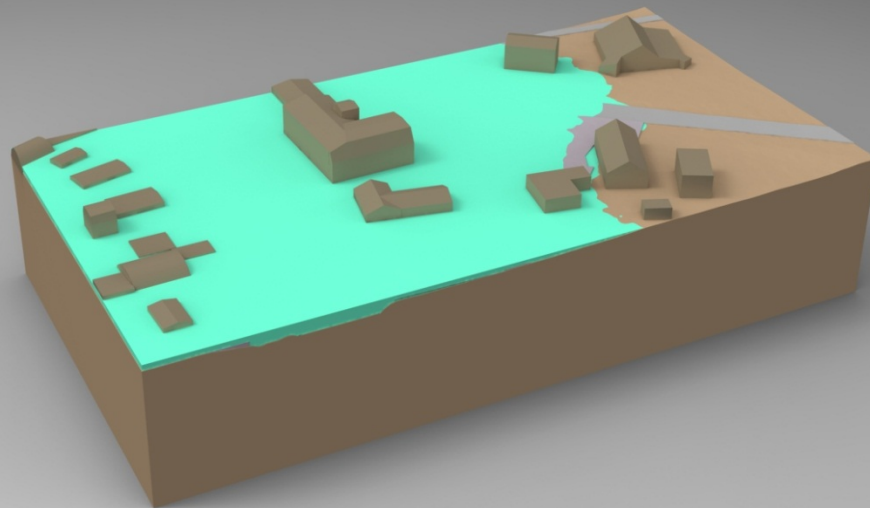
Passive Observation – 2D

Realism and limited scope

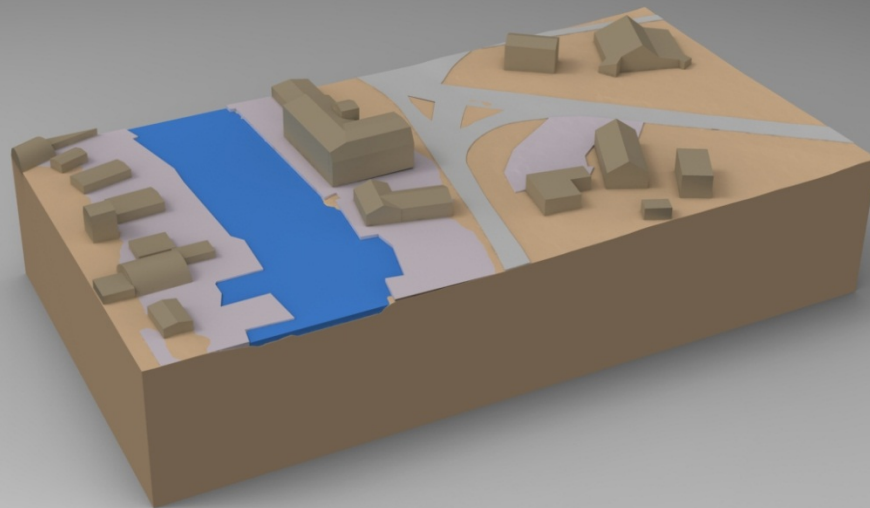


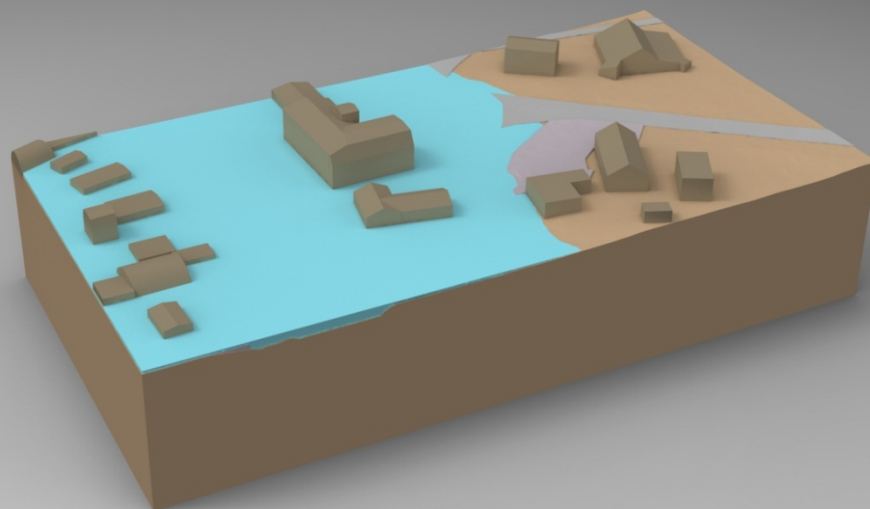
Computer dynamic models

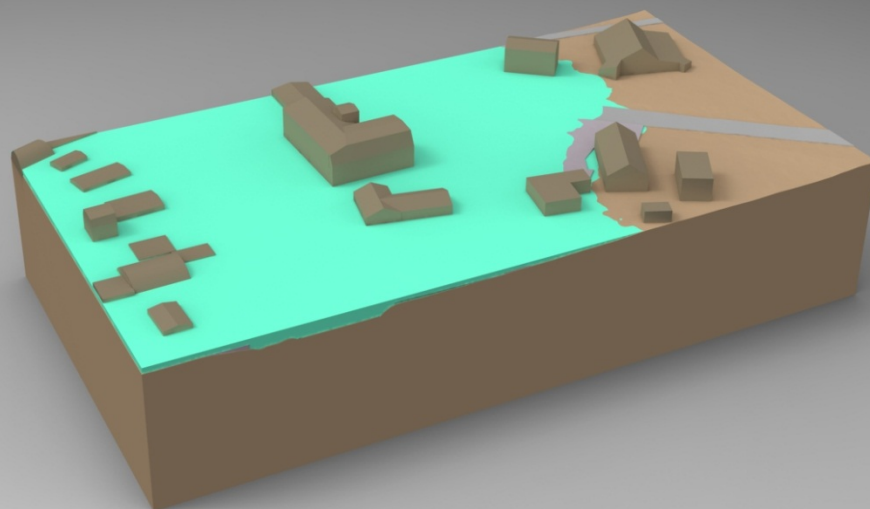
Observer and Interaction – 3D

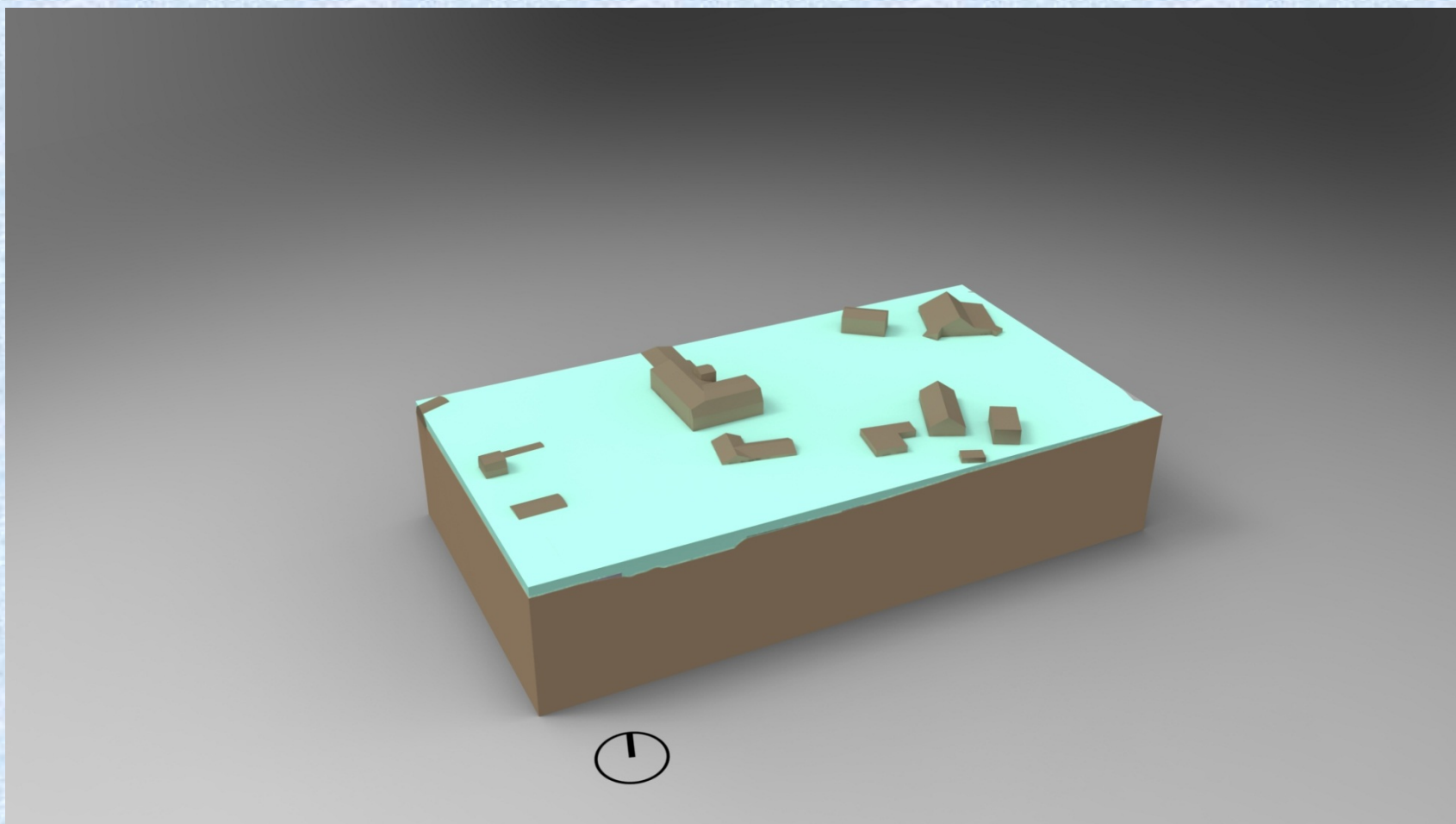


Abstraction and wide scope



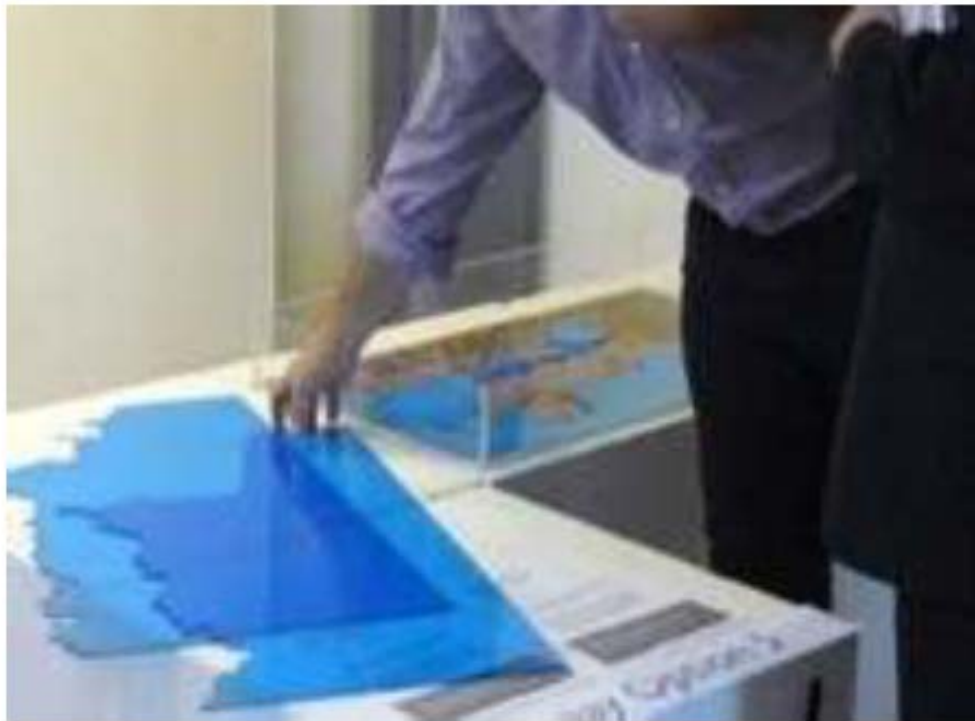






Puzzle

Observer and tactile interaction – 1D
Realism and wide scope



Flood Tank

Observer and tactile interaction -3D

Abstract and limited scope

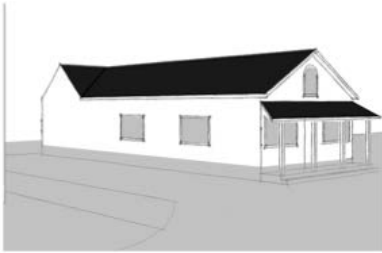


What do people think?

1. Maps
2. Rendering
3. Photo simulations
4. Computer simulation
5. Puzzle
6. Flood tank



Perception Survey



Display Option 2

This display provided clear information on how sea-level rise will affect the study area

I trust the information provided in this display on how sea-level rise will affect the study area

By viewing and/or interacting with this display my understanding on how sea-level rise will affect the study area is enhanced

I enjoyed viewing and/or interacting with this display option

comments

strongly agree

agree

disagree

strongly disagree



Display Option 6

This display provided clear information on how sea-level rise will affect the study area

I trust the information provided in this display on how sea-level rise will affect the study area

By viewing and/or interacting with this display my understanding on how sea-level rise will affect the study area is enhanced

I enjoyed viewing and/or interacting with this display option

comments

strongly agree

agree

disagree

strongly disagree

1. Maps
2. Rendering
3. Photo simulations
4. Interactive computer model
5. Puzzle
6. Flood tank

Visualisation Principals	Option 1	Option 2	Option 3	Option 4	Option 5	Option 6
Comprehension	7*	5	5	7*	9	10
Representative	6	5	4	7	9	10
Accuracy	9	8	8	8	9	10
Credibility	8	7*	7	7	9	10
Engaging	5	3	6	2	8	9
Aggregate scores	35*	28*	30	31*	44	49
Rank as determined by aggregate scores	3	6	4/5	4/5	2	1

Other work

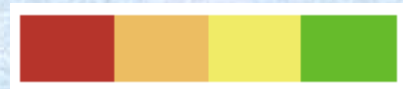
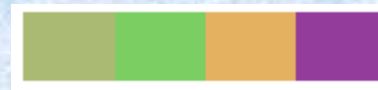
- Colour communication studies
- Communicating vulnerability 3D mentions
- Storymap

Colour and Visualisation

Colour scales

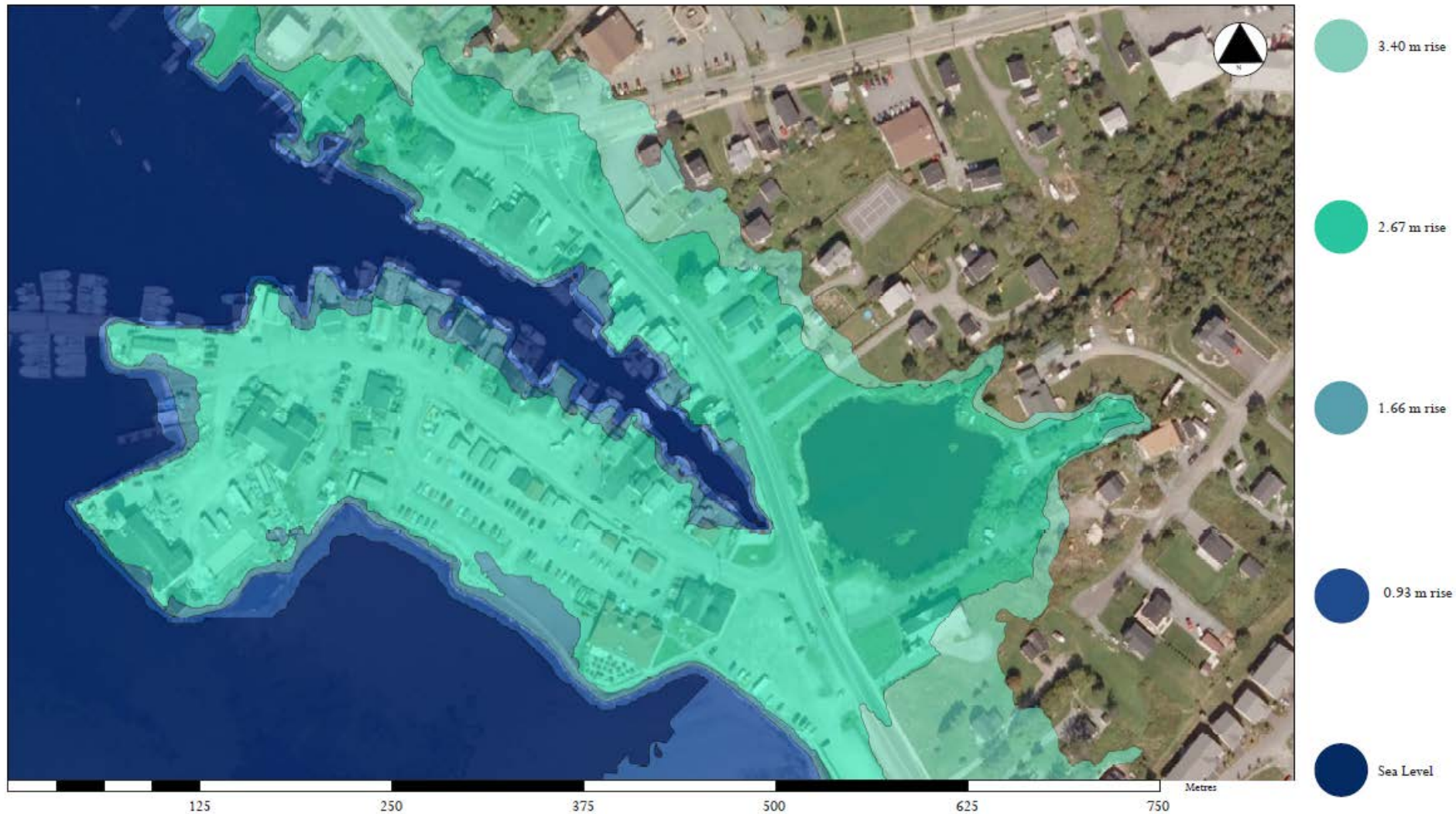
- Blue Gradient
- Red Gradient
- Contrasting Colour
- Hazard Risk

Communication



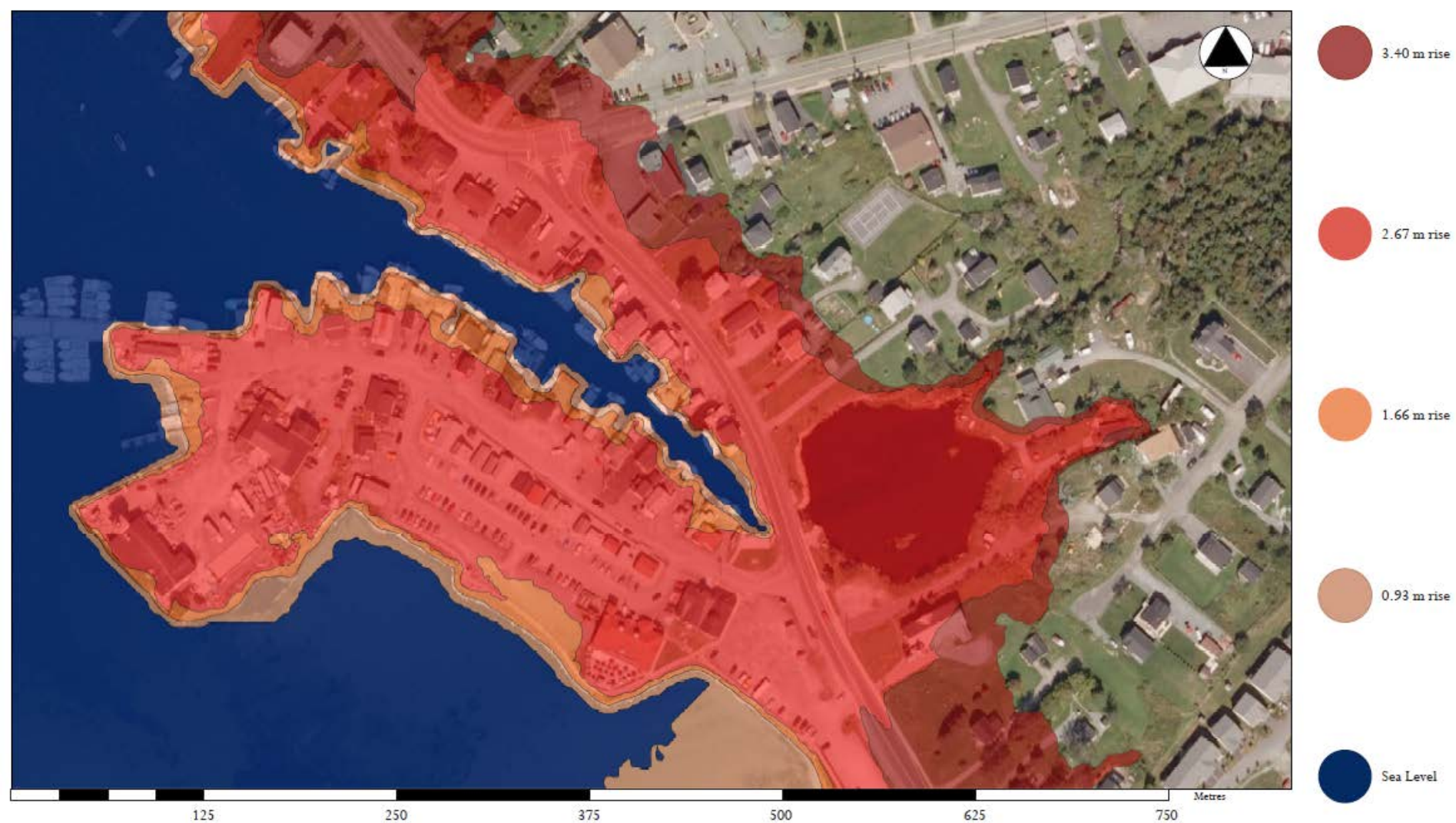
Halifax Regional Municipality Colour Scheme

Blue Gradient : Map



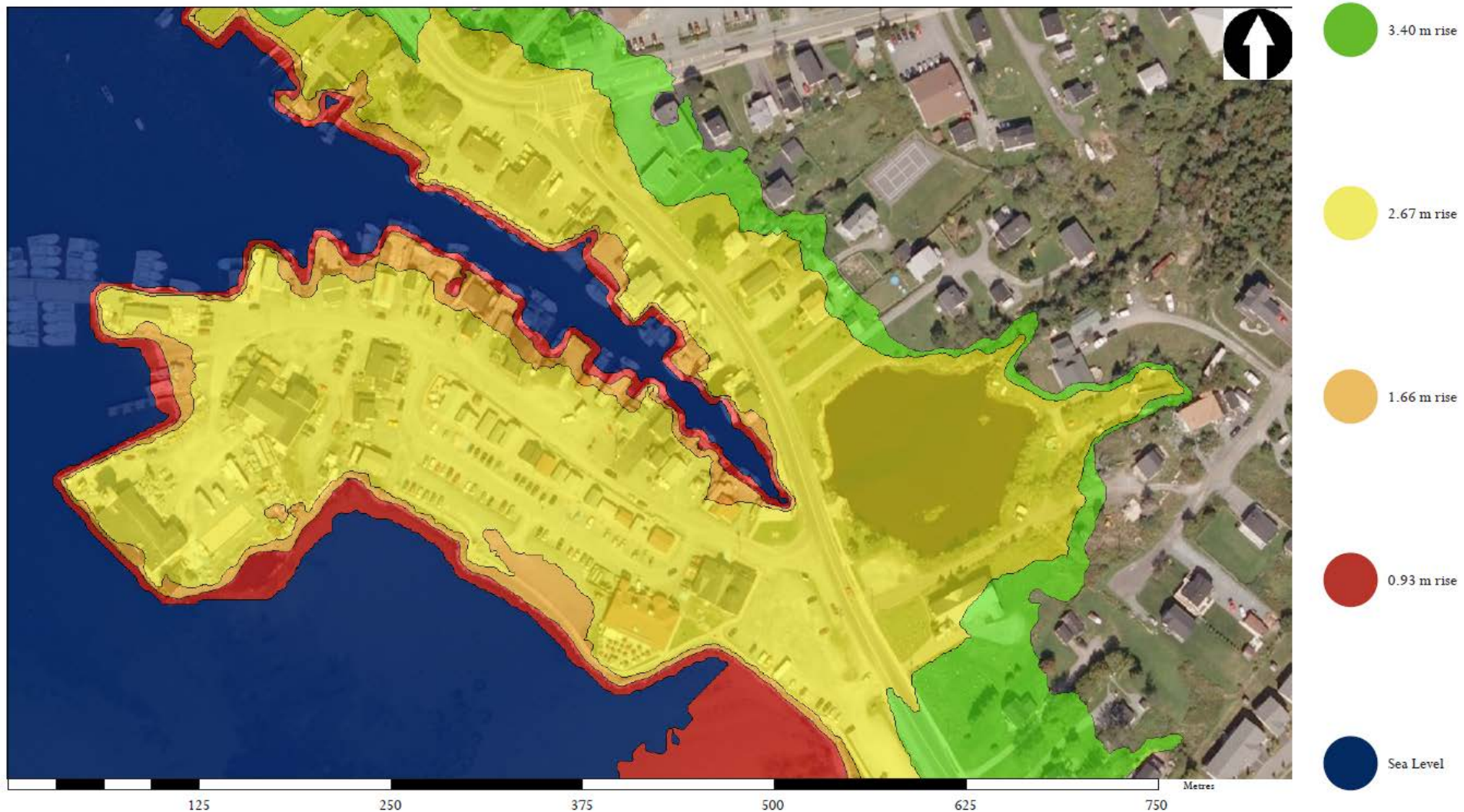
Threatening Colours Scheme

Red Gradient : Map



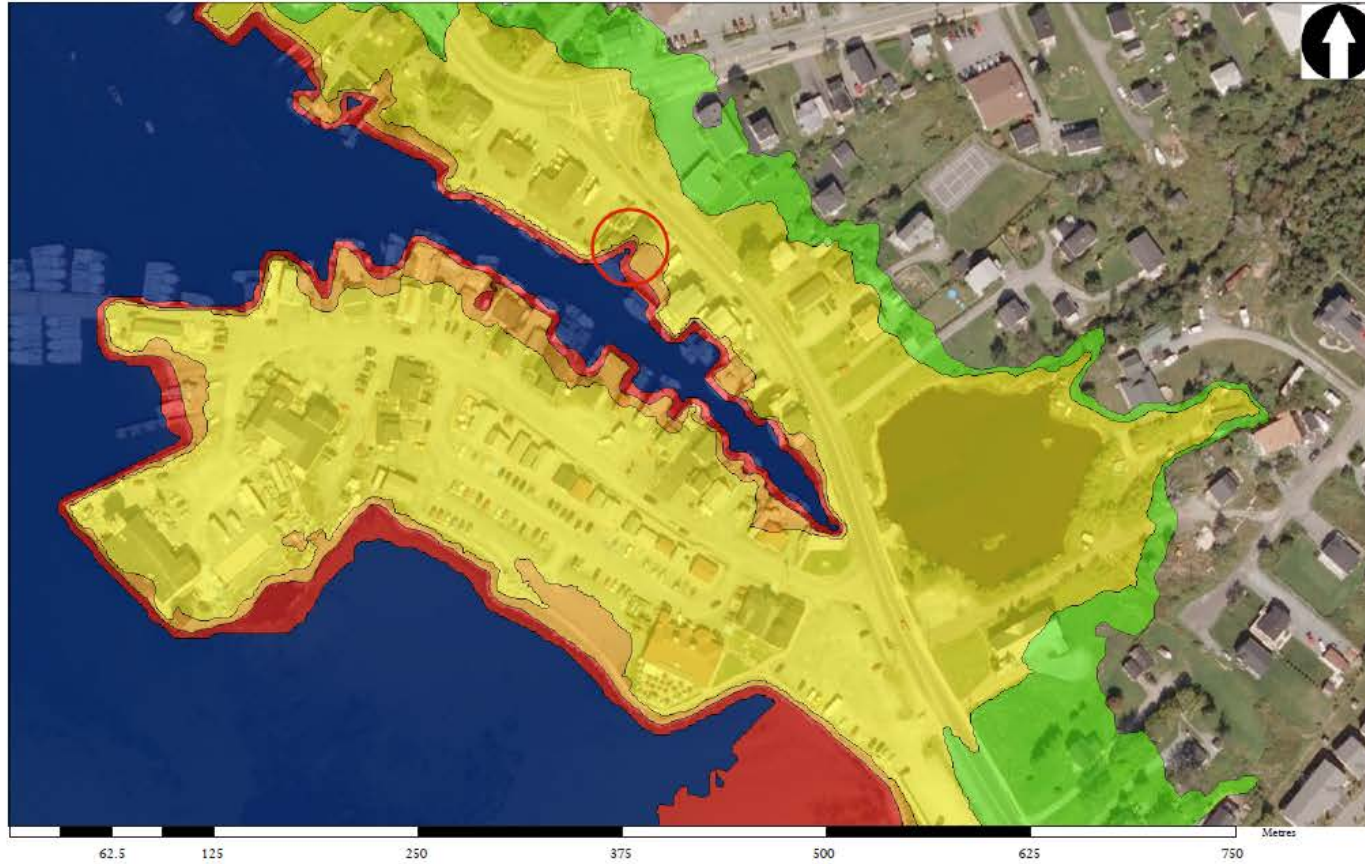
Hazard Colour Scheme

Hazard Risk Colors: Map



Add Context: Rendering

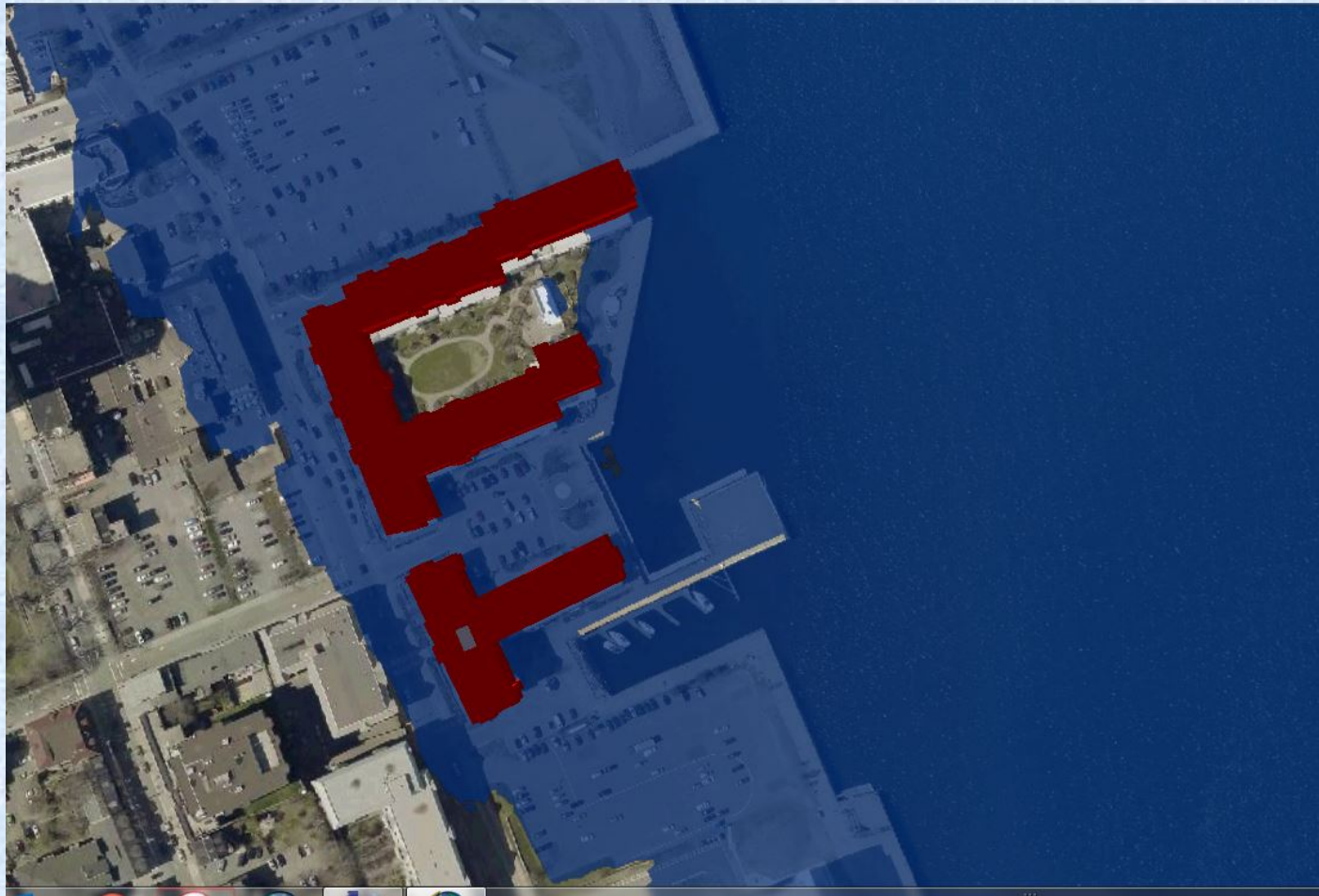
Hazard Risk Colors: Map and Rendering



Other work

- Colour communication studies
- Communicating vulnerability 3D mentions
- Storymap

Vulnerability of building by usage



Bishops Landing



Mixed use building



Vulnerability by building usage

Second floor

Condominium residential

Ground floor

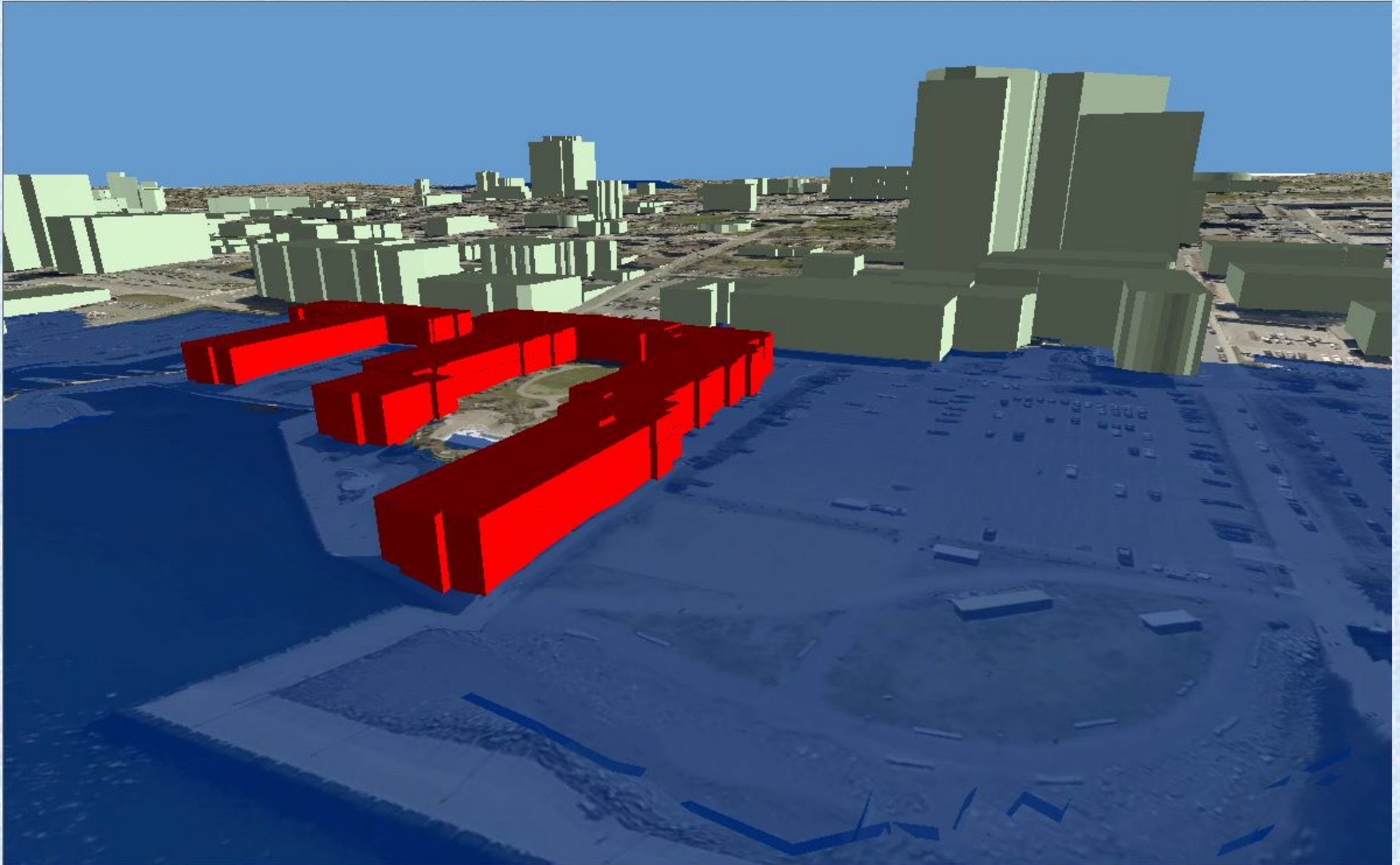
1 Dental clinic

**6 Restaurants, a wine shop, a candy shop, a hair salon,
2 clothing stores and a jewelry store , and a dry cleaner**

Ground floor and basement **indoor parking**

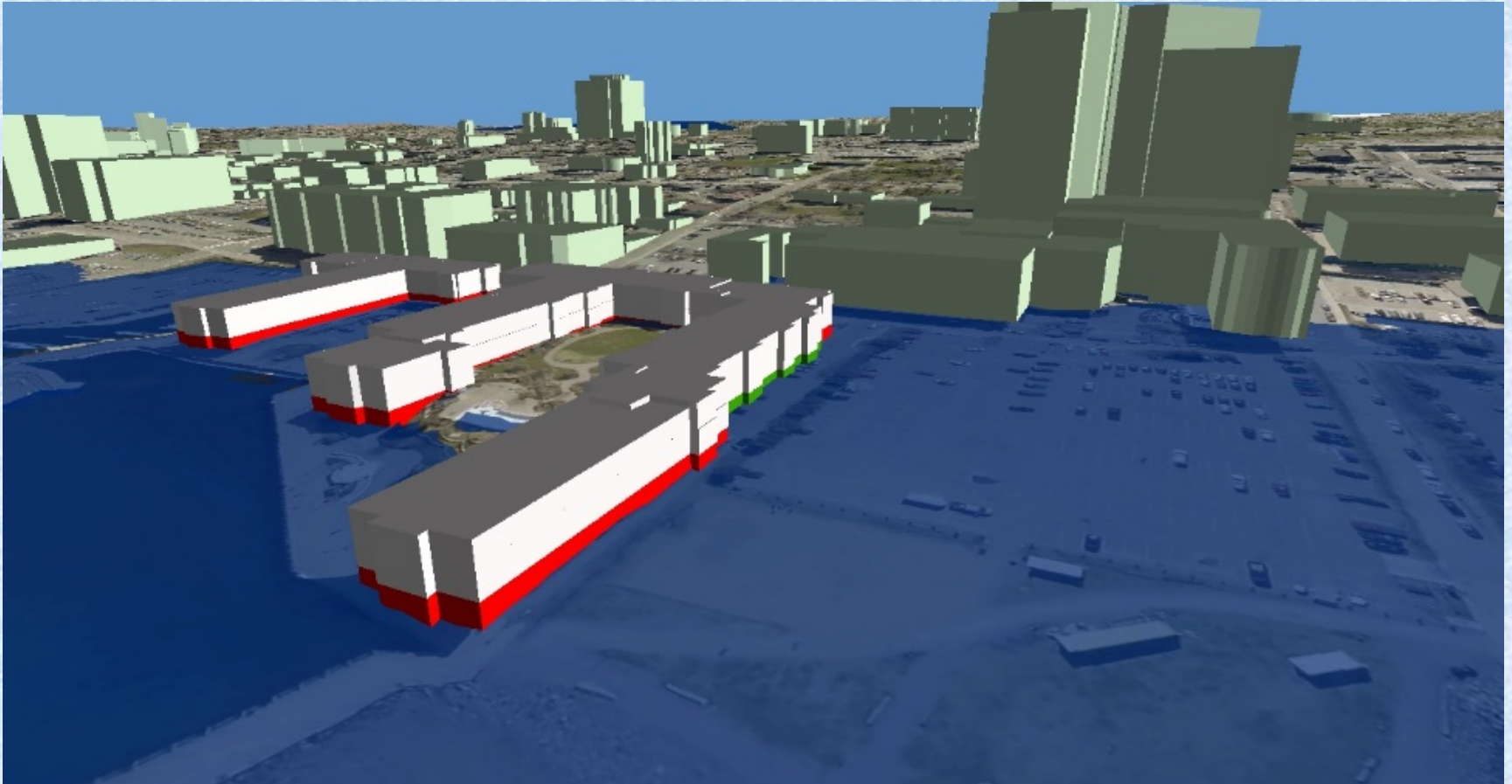
Vulnerability Assessment

Vulnerability of the different uses of the building.



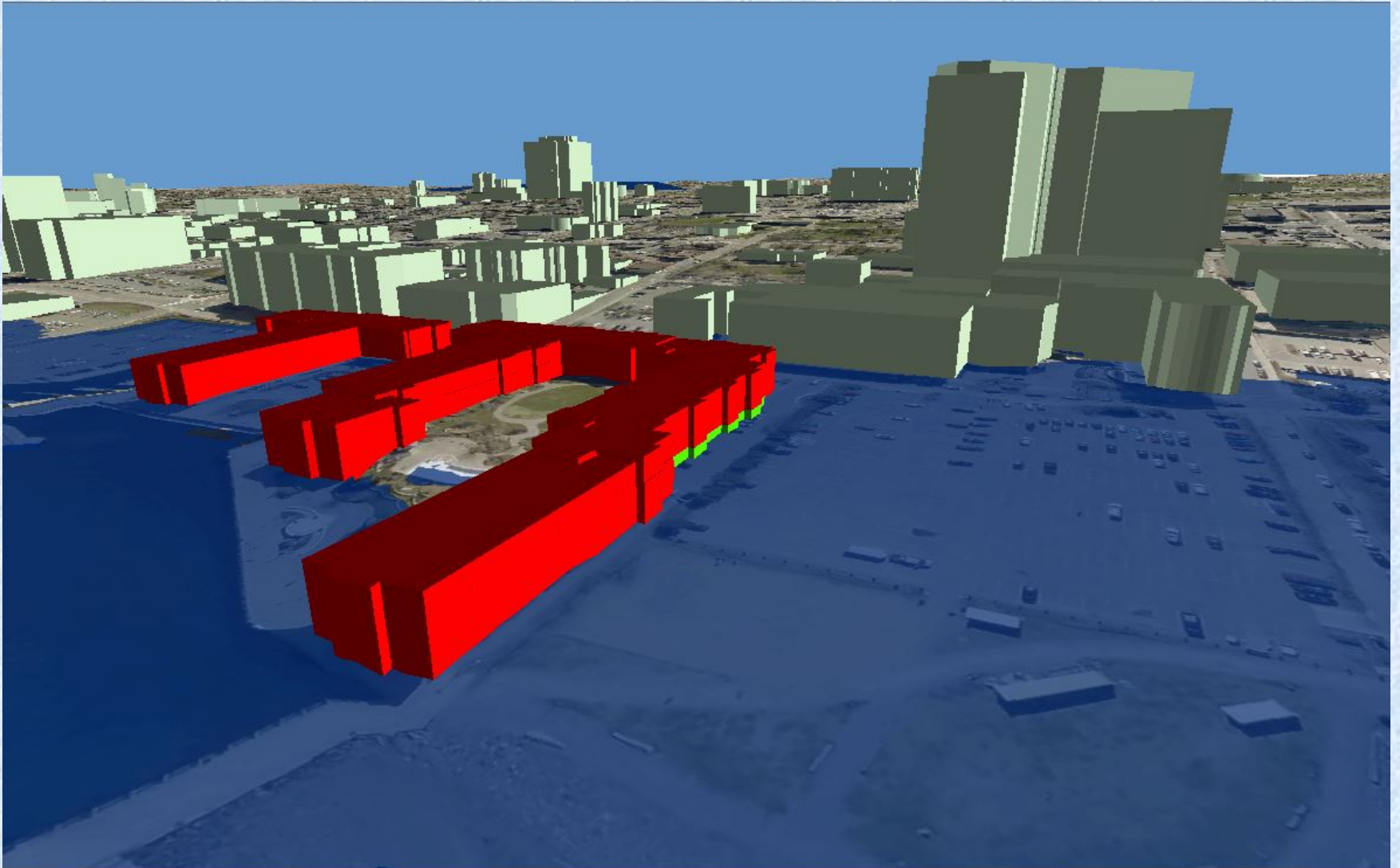
Residential and Emergency services extremely important under flooding conditions

Credibility Principle



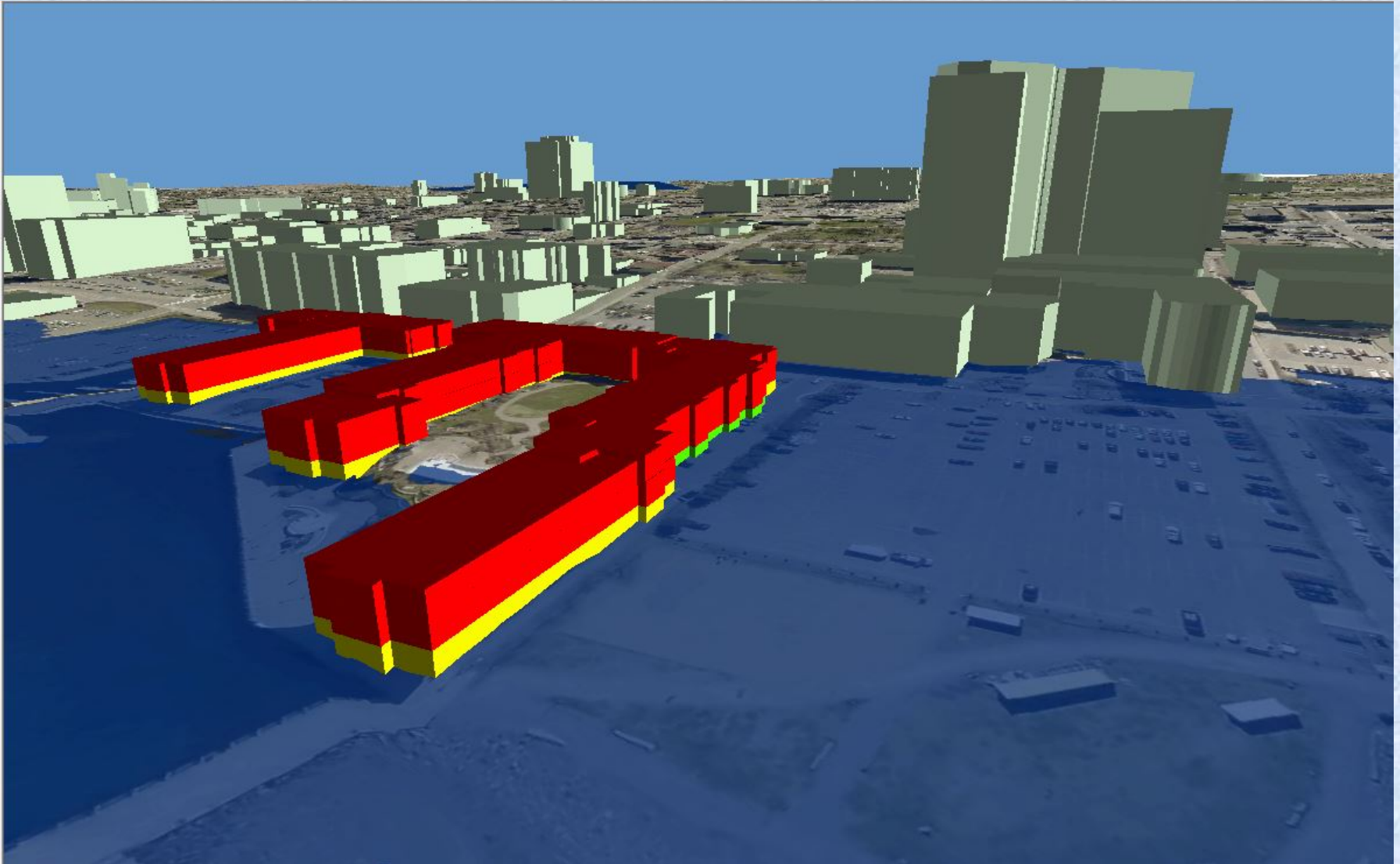
Firs story of this building is at direct risk of flooding.
Access the second floor you must get go through a lobby

Vulnerability Assessment



Garage and parking: Least vulnerable land use

Vulnerability Assessment

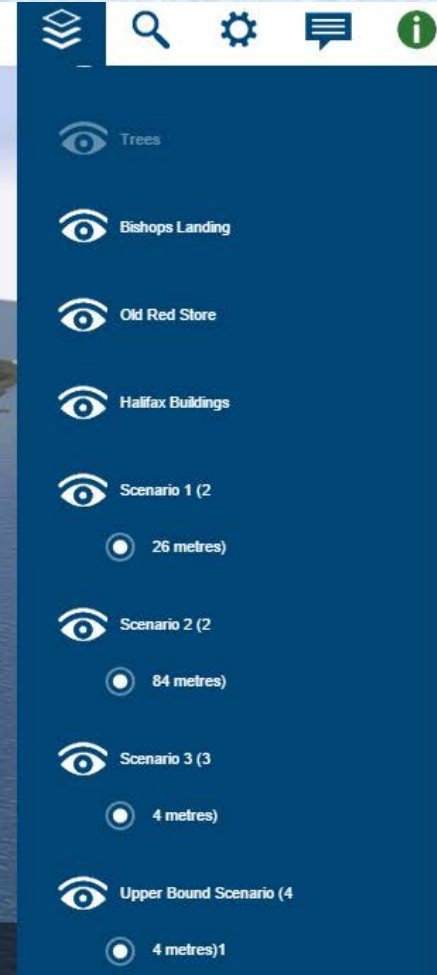
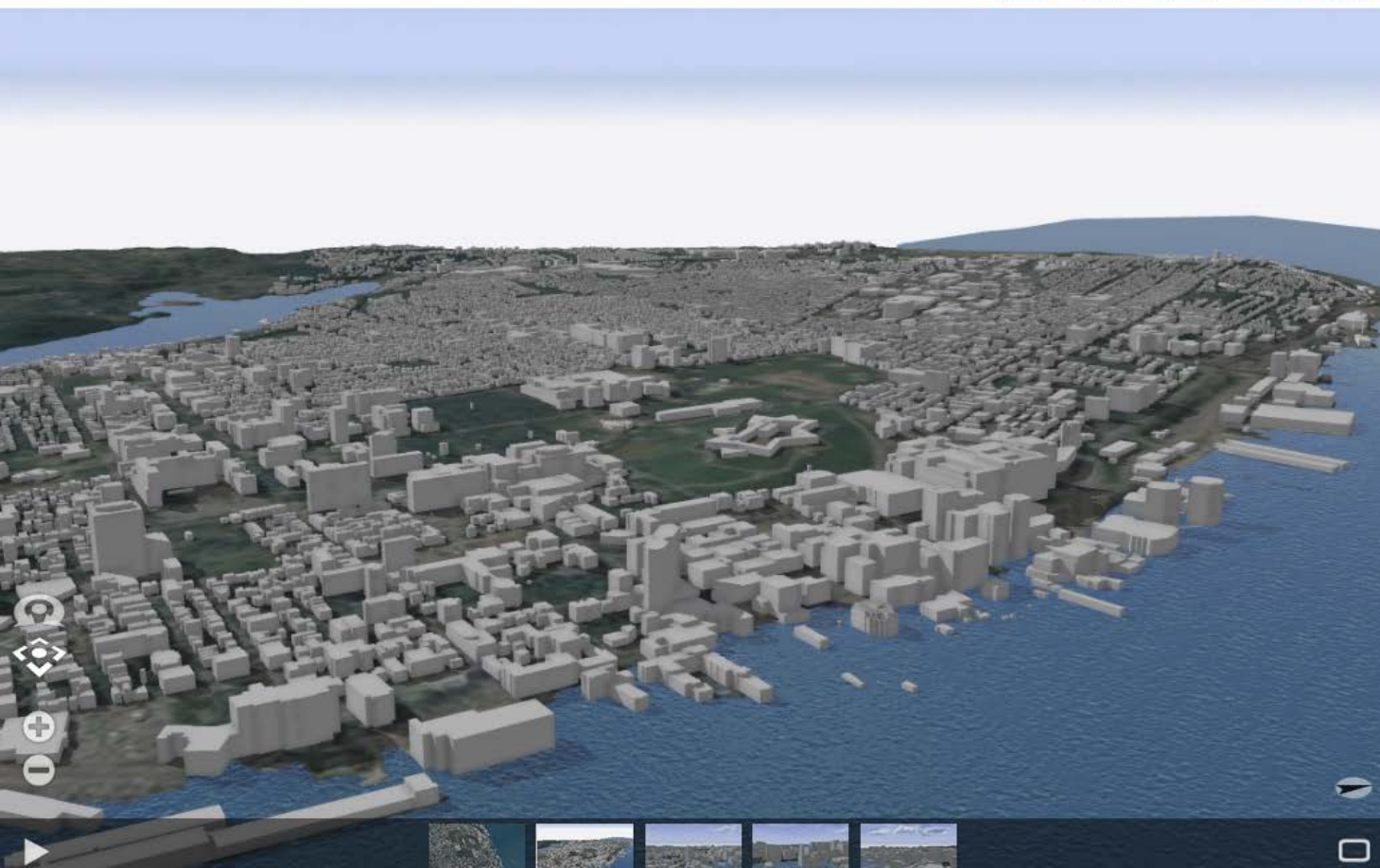


Commercial : Vulnerability depending daily needs: Clothing Store vs Grocery Store

Dynamic Visualisation Esri City Engine

Halifax Peninsula Sea Level Rise

[SHARE](#) [HELP](#) [SIGN IN](#) [ARCGIS ONLINE](#)



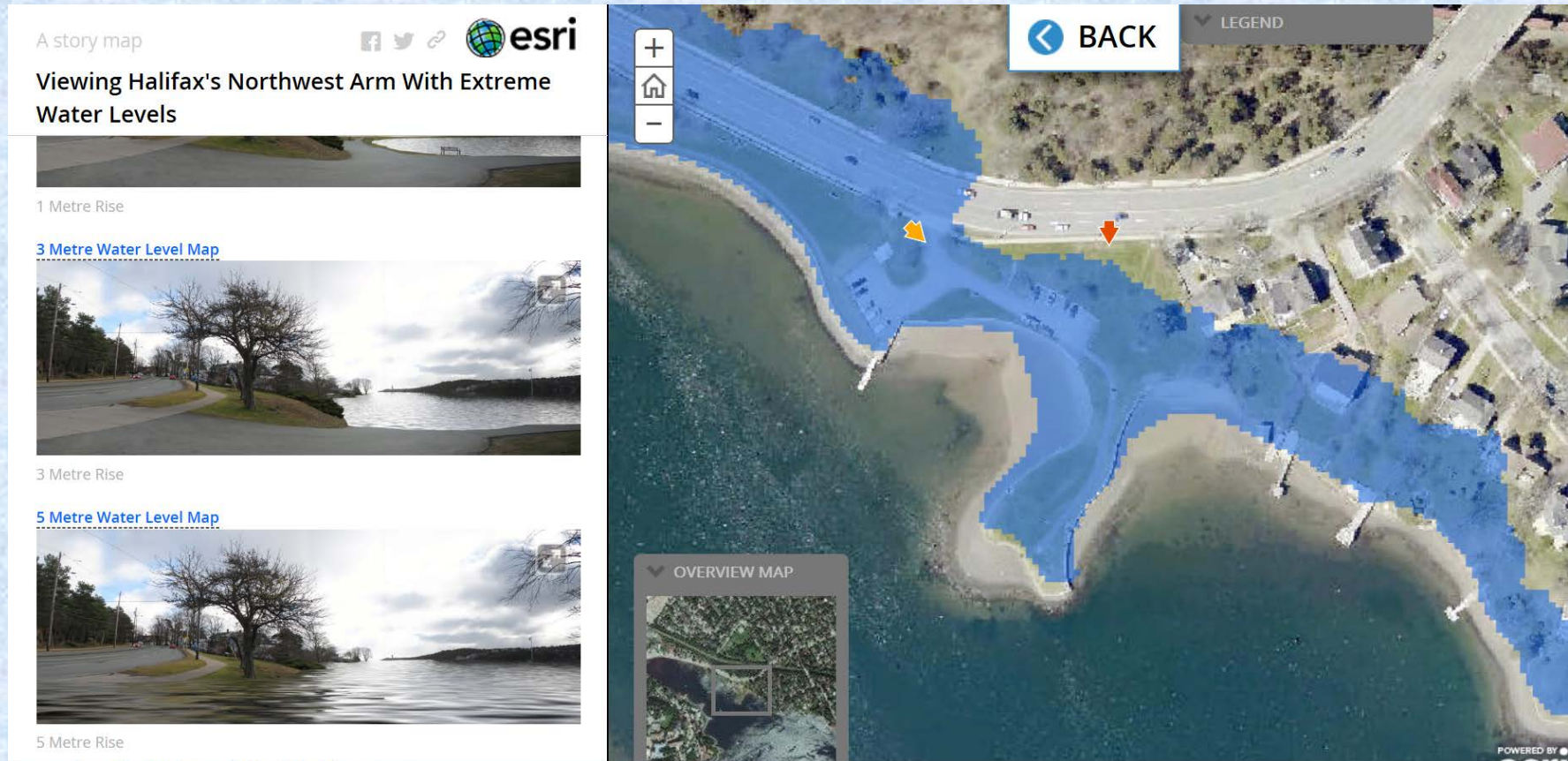
[Link to Model](#)

Other work

- Colour communication studies
- Communicating vulnerability 3D mentions
- Storymap

Esri Story maps

Allow a dynamic environment and mix of materials



A story map    

Viewing Halifax's Northwest Arm With Extreme Water Levels

Horseshoe Island Park



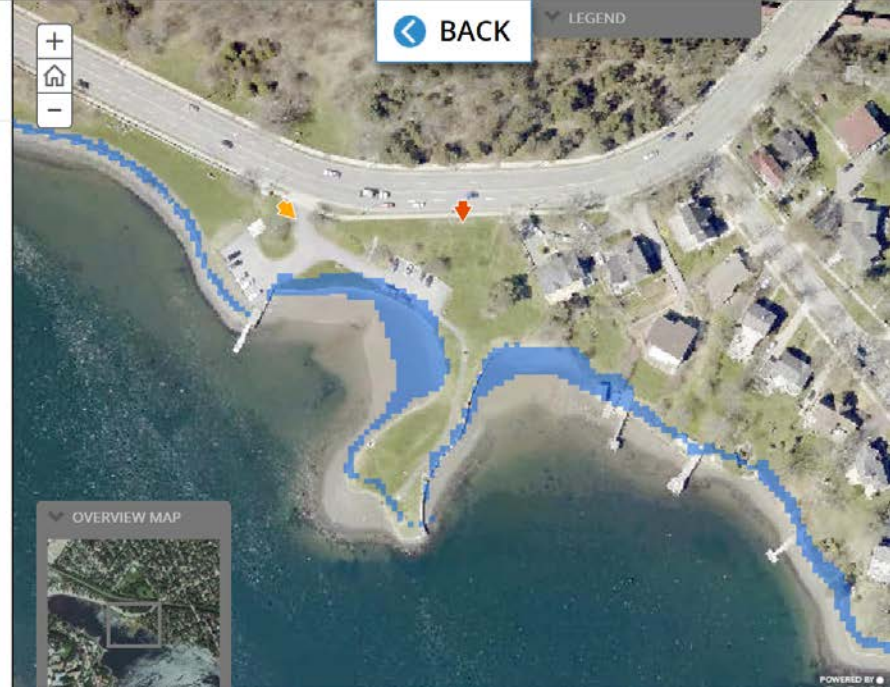
Current Condition

[1 Metre Water Level Map](#)



1 Metre Rise

[3 Metre Water Level Map](#)



A story map    

Viewing Halifax's Northwest Arm With Extreme Water Levels



1 Metre Rise

[3 Metre Water Level Map](#)



3 Metre Rise

[5 Metre Water Level Map](#)



5 Metre Rise



Challenges

- Data quality
- Multiple publics
 - Age, Education, Location
- Public accessibility to information
- Uptake by municipalities

Challenges

- Data quality
- Public accessibility
- Uptake
- Thank You
- Eric.Rapaport@dal.ca
 - Dr. Patricia Manuel
 - Paul Maher, MPS
 - Heidi Hendricks, BCD
 - Florian Goetz, BCD
- Funding Agency

Atlantic Climate Adaptation Solutions Association

Solutions d'adaptation aux changements climatiques pour l'Atlantique